

Special39 / competency practice — Electricity Paper 02 — Connections, Circuits and Diagnosis

CLASS 10 • ELECTRICITY • ORIGINAL PRACTICE

Original Class 10 Electricity • 39 questions • 80 marks • competency practice

Original chapter practice — not an official examination paper, a prediction, or an official chapter weightage.

39 questions • 80 marks • 180 minutes

Original Class X Electricity practice: circuit architecture and fault diagnosis with broad chapter coverage. Not an official CBSE paper, chapter weightage claim or examination prediction.

Paper structure

Section A | Questions 1-20 | $20 \times 1 = 20$ marks

Section B | Questions 21-26 | $6 \times 2 = 12$ marks

Section C | Questions 27-33 | $7 \times 3 = 21$ marks

Section D | Questions 34-36 | $3 \times 5 = 15$ marks

Section E | Questions 37-39 | $3 \times 4 = 12$ marks

Instructions

Attempt all questions. There are no internal OR choices. Sections A-E form the question paper. Detailed solutions and marking guidance begin only after Question 39; source-pattern and competency rationale follow in a separate appendix.

Instructions

Answer all 39 questions. There are no internal choices. Section A: $20 \times 1 = 20$ marks; B: $6 \times 2 = 12$; C: $7 \times 3 = 21$; D: $3 \times 5 = 15$; E: $3 \times 4 = 12$. Total: 80 marks. Time: 180 minutes.

Unless a question explicitly describes a fault, a changing reading, a geometry or connection change, a measurement error or a temperature change, assume steady currents, constant resistor temperature and resistance, ideal constant-voltage sources, negligible connecting-wire and closed-switch resistance, and no leakage. Ideal ammeters have zero resistance; ideal voltmeters draw no current. Open switches and broken branches do not conduct. All circuits are low-voltage conceptual models, not instructions for mains work.

Treat resistor networks as purely resistive: electrical energy dissipated in them becomes heat. Useful heat transfer or other energy losses are considered only where explicitly stated. Component power limits apply to each resistor, not merely to the network total.

Figures are original schematic diagrams, not to scale. Dots indicate conducting junctions. Follow actual connections, not proximity or shape. Source terminals shown as + and – are

connected to the stated ideal supply. For numerical answers show reasoning, units and intermediate steps.

Q1-16: choose the single correct option. Q17-20: use the common assertion-reason options listed below.

- A both true and R correctly explains A
- B both true but R does not correctly explain A
- C A true but R false
- D A false but R true

Section A

Question 1 | 1 mark

special-39-02-q01

In a metal wire in a closed circuit, conventional current is directed from point P towards point Q. In which direction is the net drift of electrons? [1 mark]

- A. From P towards Q, in the direction of conventional current
- B. From Q towards P, opposite to conventional current
- C. There is no net electron drift when current flows
- D. Electrons drift equally in both directions, giving zero net drift

Question 2 | 1 mark

special-39-02-q02

An ideal source maintaining a nonzero constant voltage is connected to a resistor and a switch in series. An ideal voltmeter remains connected directly across the source terminals. What happens to its reading when the switch changes from open to closed? [1 mark]

- A. It is zero with the switch open and nonzero when closed.
- B. It is nonzero with the switch open and zero when closed.
- C. It has the same nonzero value in both switch positions.
- D. It doubles when the switch is closed.

Question 3 | 1 mark

special-39-02-q03

In 30 s, 90 C enters a junction feeding two branches. One branch carries a steady 0.80 A. There is no charge accumulation. What current flows in the other branch? [1 mark]

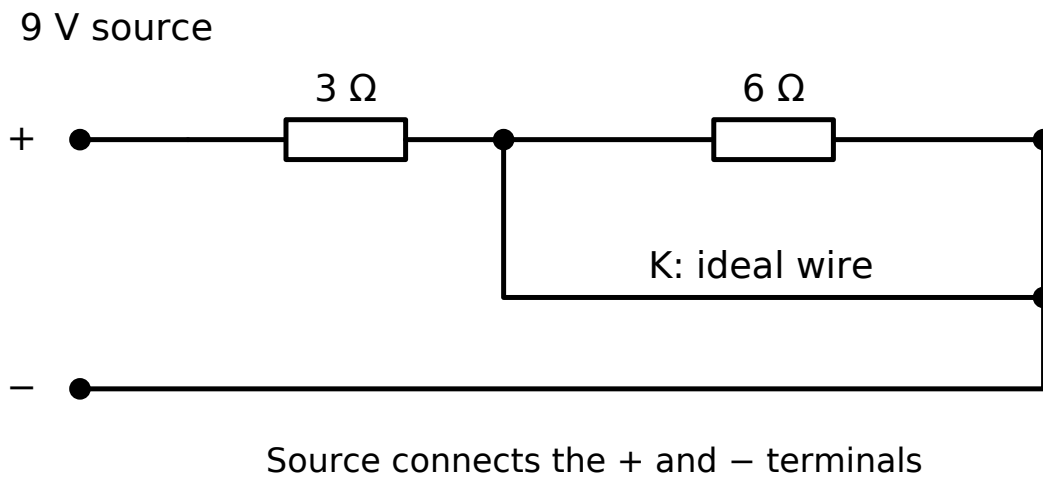
- A. 2.20 A
- B. 3.80 A
- C. 0.80 A
- D. 2.40 A

Question 4 | 1 mark

special-39-02-q04

In the supplied figure, the link K is an ideal conducting wire, not an open switch. What are the source current and the current through the $6\ \Omega$ resistor, respectively? [1 mark]

Q4 — Follow the conducting paths



- A. 1 A and 1 A
- B. 4.5 A and 1.5 A
- C. 3 A and 1.5 A
- D. 3 A and 0 A

Question 5 | 1 mark

special-39-02-q05

A 10 Ω resistor is alone across an ideal 5 V source. A 15 Ω resistor is then added in parallel. Which statement correctly describes the original resistor? [1 mark]

- A. Its current falls because current is shared equally.
- B. Its current stays 0.50 A while source current increases.
- C. Its voltage falls to 2.5 V.
- D. Its current becomes the total source current.

Question 6 | 1 mark

special-39-02-q06

At a specified temperature, which pair correctly identifies a material property and a property of a particular uniform wire specimen, respectively? [1 mark]

- A. Resistance of the wire; resistivity of its material
- B. Length of the wire; resistivity of its material
- C. Resistivity of the material; resistance of the wire
- D. Cross-sectional area of the wire; length of the wire

Question 7 | 1 mark

special-39-02-q07

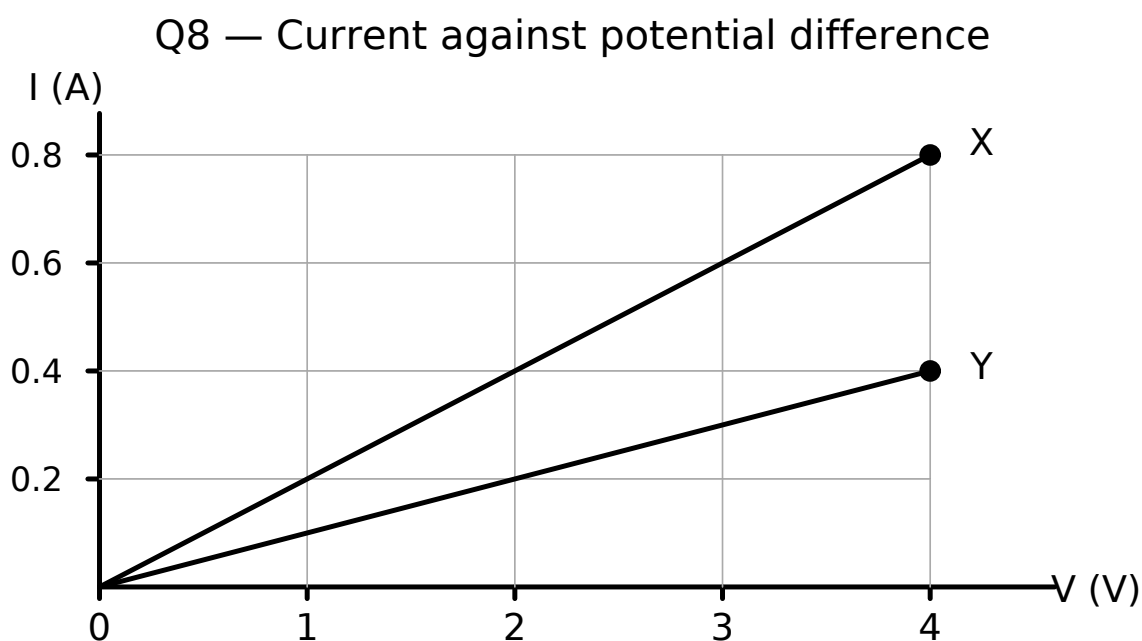
A replacement cylindrical wire has twice the length and half the diameter of the original. Material and temperature are unchanged. Its resistance is what multiple of the original resistance? [1 mark]

- A. 8
- B. 4
- C. 2
- D. $1/2$

Question 8 | 1 mark

special-39-02-q08

The figure gives the separate I-V traces of two ohmic resistors X and Y at constant temperature; I is vertical and V is horizontal. Both resistors are then used once, either in series or in parallel, across an adjustable ideal source. A proposed trace for the combined network is a straight line through the origin and the point ($V = 3.0 \text{ V}$, $I = 0.90 \text{ A}$), where I is the total source current. Which conclusion correctly tests the proposed trace against the figure? [1 mark]



- A. The series connection fits: at 3.0 V the total current is 0.90 A .
- B. The series connection fits: at 3.0 V the total current is 0.60 A .
- C. The parallel connection fits: at 3.0 V the total current is 0.20 A .
- D. The parallel connection fits: at 3.0 V the total current is 0.90 A .

Question 9 | 1 mark

special-39-02-q09

Resistors $4\ \Omega$ and $8\ \Omega$ are in series. An ideal voltmeter reads 4 V across the $4\ \Omega$ resistor. Without changing the circuit, it is moved to span both resistors. What will it read? [1 mark]

- A. 4 V
- B. 8 V
- C. 12 V
- D. 16 V

Question 10 | 1 mark

special-39-02-q10

Which statement correctly describes one kilowatt-hour? [1 mark]

- A. A power of 1000 W per hour
- B. An energy of $3.6 \times 10^6\ \text{J}$
- C. A current of 1000 A in one hour
- D. An energy of 3600 J

Question 11 | 1 mark

special-39-02-q11

A resistor R is replaced by a resistor $2R$ across the same ideal voltage source for the same time. What happens to the heat generated? [1 mark]

- A. It halves.
- B. It doubles.
- C. It quadruples.
- D. It stays unchanged.

Question 12 | 1 mark

special-39-02-q12

Three separate resistors are connected in parallel directly across an ideal source. One develops a complete break internally. What happens to the other two steady branch currents? [1 mark]

- A. Both become zero.
- B. Both double.
- C. They become equal regardless of resistance.
- D. Both remain unchanged.

Question 13 | 1 mark

special-39-02-q13

A $2\ \Omega$ resistor rated at a maximum $2\ \text{W}$ and an $8\ \Omega$ resistor rated at a maximum $8\ \text{W}$ are in series. Ratings may be reached but not exceeded. Which is the largest source voltage that is allowed for both? [1 mark]

- A. $4\ \text{V}$
- B. $8\ \text{V}$
- C. $10\ \text{V}$
- D. $12\ \text{V}$

Question 14 | 1 mark

special-39-02-q14

A device consumes a constant $50\ \text{W}$ for $2\ \text{h}$. How much electrical energy does it consume? [1 mark]

- A. $0.025\ \text{kWh}$
- B. $0.10\ \text{kWh}$
- C. $25\ \text{kWh}$
- D. $100\ \text{kWh}$

Question 15 | 1 mark

special-39-02-q15

In a wire experiment the measured voltage remains $6.0\ \text{V}$, but the current falls from $0.60\ \text{A}$ immediately after connection to $0.50\ \text{A}$ later. Which conclusion is supported by these readings alone? [1 mark]

- A. The wire resistance increased.
- B. The source voltage decreased.
- C. The charge of each electron increased.
- D. The readings prove the wire became colder.

Question 16 | 1 mark

special-39-02-q16

Two model lamps rated $12\ \text{V}$, $6\ \text{W}$ and $12\ \text{V}$, $12\ \text{W}$ are connected in series across $12\ \text{V}$. Treat their resistances as constant at their rated values and brightness as proportional to electrical power. Which is brighter? [1 mark]

- A. The $12\ \text{W}$ lamp, because its rated power is higher.
- B. They are equally bright because their current is equal.
- C. Neither takes any current.
- D. The $6\ \text{W}$ lamp, because its resistance is greater.

Question 17 | 1 mark

special-39-02-q17

Assertion (A): In a steady series circuit, the current before and after a resistor is the same.

Reason (R): A series circuit has no branch path and charge does not accumulate in the resistor. [1 mark]

- A. both true and R correctly explains A
- B. both true but R does not correctly explain A
- C. A true but R false
- D. A false but R true

Question 18 | 1 mark

special-39-02-q18

Assertion (A): Cutting a uniform wire into two equal lengths halves the resistivity of each piece at the same temperature. Reason (R): Resistivity is a characteristic of the material at a specified temperature. [1 mark]

- A. both true and R correctly explains A
- B. both true but R does not correctly explain A
- C. A true but R false
- D. A false but R true

Question 19 | 1 mark

special-39-02-q19

Assertion (A): Two unequal resistors connected in parallel have the same potential difference.

Reason (R): One volt equals one joule per coulomb. [1 mark]

- A. both true and R correctly explains A
- B. both true but R does not correctly explain A
- C. A true but R false
- D. A false but R true

Question 20 | 1 mark

special-39-02-q20

Assertion (A): Doubling a resistor's resistance while keeping its voltage fixed doubles its power.

Reason (R): For a resistor, $P = V^2/R$. [1 mark]

- A. both true and R correctly explains A
- B. both true but R does not correctly explain A
- C. A true but R false
- D. A false but R true

Section B

Question 21 | 2 marks

special-39-02-q21

During the same 40 s interval, 64 C passes through the supply lead of a two-branch circuit and 24 C passes through branch X. A student says branch Y must carry the same current as X because both are parallel. (i) Determine the current in Y. (1 mark) (ii) Decide whether the student's claim is valid, explaining the governing relationship. (1 mark)

Question 22 | 2 marks

special-39-02-q22

A charge of 15 C passes through a resistor across which the potential difference is 8 V. (i) State what a potential difference of 8 V means in terms of energy and charge. (1 mark) (ii) Calculate the electrical energy transferred to the resistor. (1 mark)

Question 23 | 2 marks

special-39-02-q23

A current meter consistently displays +0.04 A with no current. It then displays 0.34 A while an ideal voltmeter reads 3.6 V across a resistor. Apart from this additive zero offset, the current meter is ideal. A report gives the resistance as $3.6/0.34 \approx 10.6 \, \Omega$. (i) Find the corrected resistance. (1 mark) (ii) Explain whether the reported value is too high or too low, without merely quoting your result. (1 mark)

Question 24 | 2 marks

special-39-02-q24

A student tests whether wire resistance depends on length by comparing a short thin copper wire with a long thick alloy wire at the same voltage. (i) Explain why this test cannot isolate the effect of length. (1 mark) (ii) Specify a corrected pair of wires and one operating precaution needed for the comparison. (1 mark)

Question 25 | 2 marks

special-39-02-q25

A uniform wire is 1.5 m long, has cross-sectional area $0.30 \, \text{mm}^2$ and resistance $2.0 \, \Omega$. (i) Express its area in m^2 . (1 mark) (ii) Calculate the resistivity of its material at the test temperature. (1 mark)

Question 26 | 2 marks

special-39-02-q26

Two ammeters in a single steady series loop read the same current, one just before and one just after a heater. The heater nevertheless transfers electrical energy to its surroundings. (i) Use the readings to assess the claim "the heater uses up electric charge". (1 mark) (ii) Explain what the source supplies and what changes as charge passes through the heater. (1 mark)

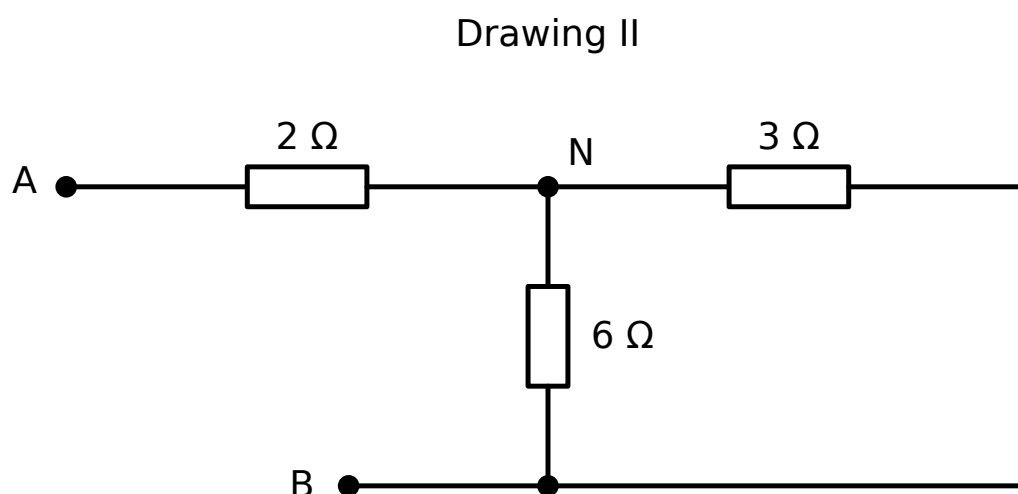
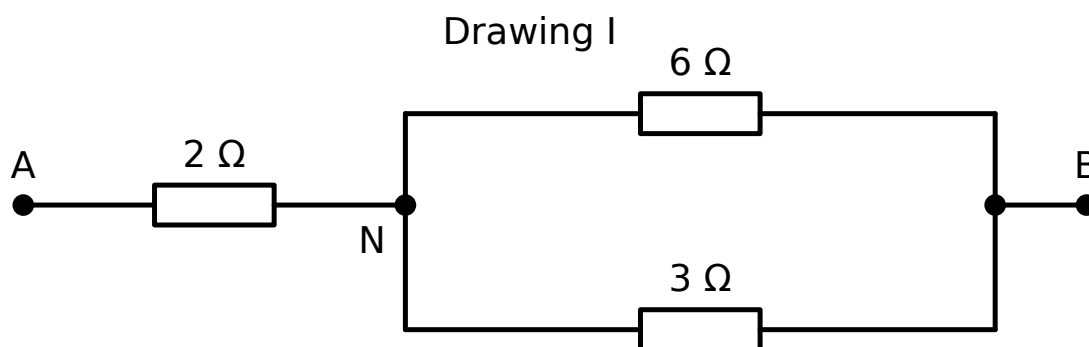
Section C

Question 27 | 3 marks

special-39-02-q27

Drawings I and II in the figure use the same three resistor values and are each connected across a separate ideal 8 V source at terminals A and B. Dots denote joins. A student treats drawing II as three resistors in series because its components are spread around the page. (i) Identify the two endpoints of each resistor in each drawing, using node N where shown. (1 mark) (ii) Find the equivalent resistance in each drawing. (1 mark) (iii) Decide whether the sources deliver the same current and justify your answer numerically. (1 mark)

Q27 — Two circuit drawings



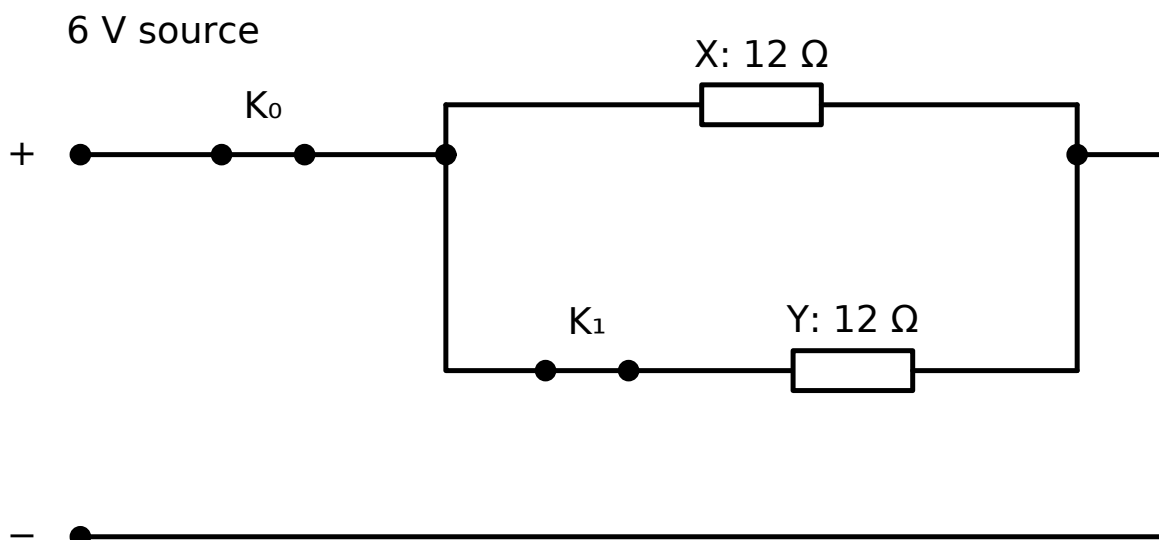
Each A-B pair connects to its own 8 V source.

Question 28 | 3 marks

special-39-02-q28

Use the switch positions and resistor values in the supplied figure. The ideal source is 6 V. (i) With both K_0 and K_1 closed, determine the source current. (1 mark) (ii) Keep K_0 closed but open K_1 : determine the source current and state what happens to the current in resistor X, explaining why. (1 mark) (iii) Close K_1 and open K_0 : determine the source current and explain why the effect differs from opening K_1 alone. (1 mark)

Q28 — Switch locations matter



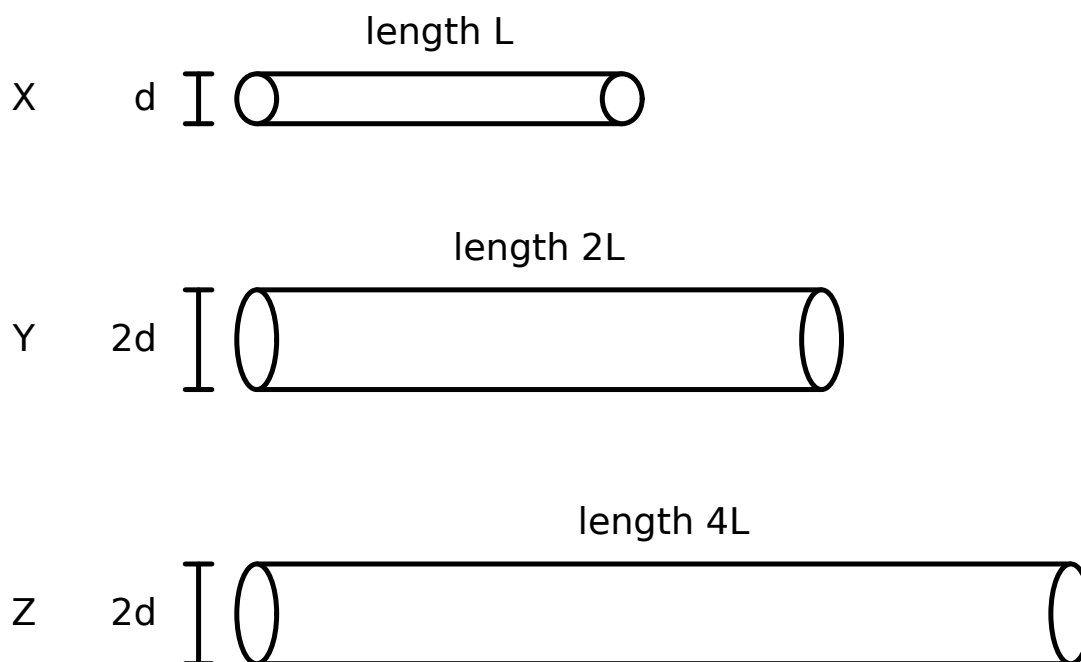
Both switches are drawn closed.

Question 29 | 3 marks

special-39-02-q29

The figure shows three uniform cylindrical wires of the same material at the same temperature. Their marked dimensions, not the apparent drawing sizes, are authoritative. (i) Find $R_X : R_Y : R_Z$. (1 mark) (ii) Select the two wires which give equal current when separately connected to the same ideal voltage source, with a reason. (1 mark) (iii) Can all three have the same resistivity despite the result in (i)? Explain. (1 mark)

Q29 — Dimensions of wire samples



d and $2d$ label diameters, not areas.

All wires: same material and temperature; not to scale.

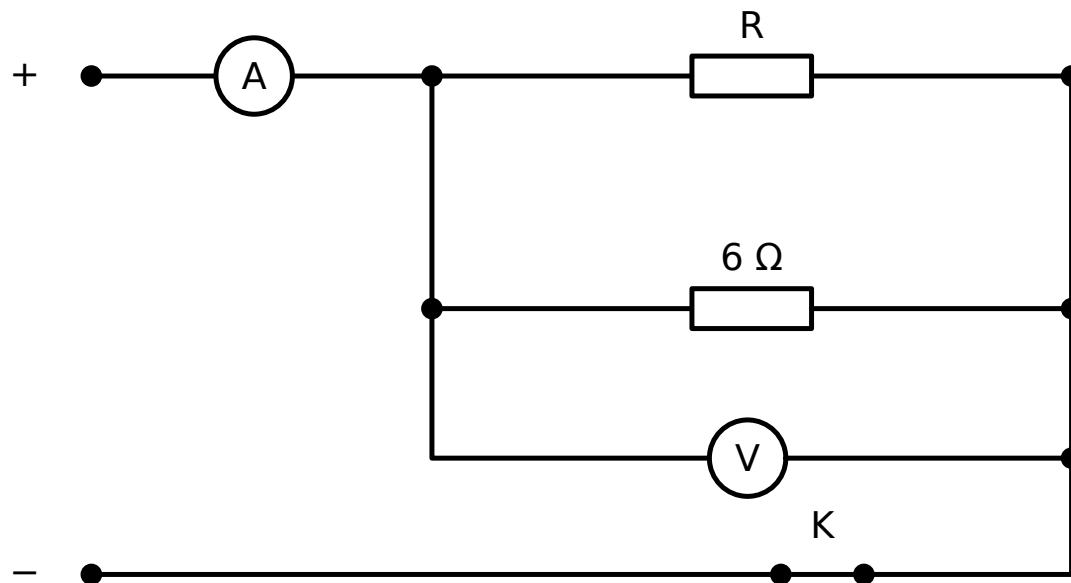
Question 30 | 3 marks

special-39-02-q30

The figure is an attempted resistance-measurement circuit. Its ideal voltmeter reads 6.0 V and its ideal ammeter reads 1.5 A with the switch closed. A report claims that the resistor labelled R is therefore $4\ \Omega$. (i) Explain the connection error that invalidates the claim. (1 mark) (ii) Use the readings to determine R correctly. (1 mark) (iii) State how to reconnect the ammeter to measure only the current through R while retaining the voltmeter position. (1 mark)

Q30 — Inspect the meter connections

Ideal 6 V source



A: 1.5 A V: 6.0 V

Question 31 | 3 marks

special-39-02-q31

Two resistors of $3\ \Omega$ and $6\ \Omega$ are tested for 60 s. (i) When connected in series, find the ratio $H_{3\Omega} : H_{6\Omega}$ of heat produced, explaining the controlled quantity. (1 mark) (ii) When connected in parallel across the same ideal source, find that ratio and explain why it reverses. (1 mark) (iii) State Joule's heating formula in terms of current, resistance and time. (1 mark)

Question 32 | 3 marks

special-39-02-q32

A display must remain on continuously for 4 h. Plan A uses a 30 W lamp for all 4 h. Plan B uses an 18 W lamp for the first 3 h and a 42 W lamp for the final 1 h. All operate at their stated powers; both plans meet the lighting requirement. (i) Calculate the energy for each plan in kWh. (1 mark) (ii) A budget allows at most 0.10 kWh. Select the feasible plan and state its energy margin below the budget. (1 mark) (iii) At ₹8 per kWh, find the cost saving of that plan relative to the other. (1 mark)

Question 33 | 3 marks

special-39-02-q33

An ideal 12 V supply feeds three parallel branches of $6\ \Omega$, $12\ \Omega$ and $24\ \Omega$. A fault makes exactly one entire branch open; all other components stay unchanged. The total current falls from 3.5 A to 2.5 A. (i) Identify the open branch using the size of the change. (1 mark) (ii) Predict the currents in the two intact branches. (1 mark) (iii) An ideal voltmeter is placed across the terminals of the broken branch. Predict its reading and explain why zero current does not imply

zero voltage here. (1 mark)

Section D

Question 34 | 5 marks

special-39-02-q34

A test load must draw exactly 1.0 A from an ideal 12 V source. You must use all four available identical $12\ \Omega$ resistors, each rated for a maximum continuous power of 4.0 W. Only series and parallel connections are allowed; do not bypass a resistor. (i) Determine the required equivalent resistance. (1 mark) (ii) Design a suitable network: describe every branch connection clearly or draw a labelled circuit, and verify its equivalent resistance. (2 marks) (iii) Verify the power dissipated by each resistor and decide whether the design is acceptable. (1 mark) (iv) For your unchanged design, find the largest supply voltage permitted by the individual 4.0 W ratings. Give an exact or suitably rounded value. (1 mark)

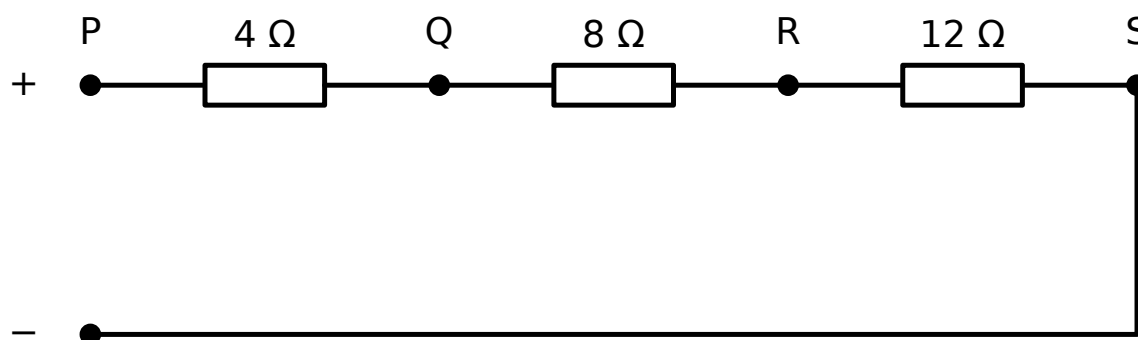
Question 35 | 5 marks

special-39-02-q35

The figure shows a series test chain connected across an ideal 12 V source. Only the three labelled resistors can be faulty; wires, source and ideal meters are sound. Test 1: the source current is zero. Voltage measurements are $V_{PQ} = 0\text{ V}$, $V_{QR} = 12\text{ V}$ and $V_{RS} = 0\text{ V}$ (magnitudes). Exactly one resistor has become a complete open circuit. (i) Identify it and explain all three readings. (2 marks) (ii) After replacing it with an intact resistor of its labelled value, calculate the source current and the voltage across the $8\ \Omega$ resistor. (1 mark) Test 2 is a separate fault after full repair: an accidental ideal wire bypasses exactly one resistor and source current becomes 1.0 A. (iii) Identify the bypassed resistor using equivalent resistance, and predict its voltage. (2 marks)

Q35 — Series test chain

12 V source



Measure voltage between the labelled terminals.

Question 36 | 5 marks

special-39-02-q36

A small resistive warming plate can use either mode F, one $12\ \Omega$ element across $12\ \text{V}$, or mode S, two identical $12\ \Omega$ elements in series across $12\ \text{V}$. The element resistances remain constant. All electrical energy generated in the active elements reaches the plate; ignore other losses. The plate needs $720\ \text{J}$ within $90\ \text{s}$. (i) Calculate the total power in each mode. (2 marks) (ii) Determine which mode can meet the deadline, using its heating time. (1 mark) (iii) A student claims the lower-power mode necessarily uses less electrical energy to deliver the same $720\ \text{J}$. Evaluate the claim and find the charge delivered by the source for a complete delivery of $720\ \text{J}$ in each mode, allowing mode S to run beyond the deadline. (2 marks)

Section E

Question 37 | 4 marks

special-39-02-q37

A technician receives a sealed two-terminal box containing exactly two intact $18\ \Omega$ resistors, connected either in series or in parallel. An ideal source and ideal ammeter give these steady test records at constant temperature: $3.0\ \text{V} \rightarrow 0.333\ldots\ \text{A}$; $6.0\ \text{V} \rightarrow 0.666\ldots\ \text{A}$. Later, at the same $6.0\ \text{V}$, the current becomes $0.333\ldots\ \text{A}$. Exactly one internal resistor has become a complete open circuit; no other connection has changed. (i) Infer the original arrangement from both initial readings. (1 mark) (ii) Explain the later reading by calculating the new resistance and relating it to the stated failure. (1 mark) (iii) After the fault, the technician proposes doubling the voltage to restore the former total current. If each resistor has a $5\ \text{W}$ maximum power rating, assess this proposal quantitatively. (2 marks)

Question 38 | 4 marks

special-39-02-q38

A club builds a low-voltage sign from two resistive strips connected as parallel branches across an ideal $10\ \text{V}$ source. Strip A has resistance $20\ \Omega$ and strip B $50\ \Omega$. Each branch has its own ideal series switch, and there is a master switch in the common supply lead. In a trial, both strips operate for the first 4 min; only A operates for the next 2 min; then the master switch is opened. All resistances remain constant. A log states that opening B's switch should reduce A's current, and also claims that total source energy for the 6 min was $1800\ \text{J}$. (i) Assess the current claim with a value and reason. (1 mark) (ii) Calculate total source energy and decide whether the energy log is correct. (2 marks) (iii) Explain why opening the master switch, unlike opening B's branch switch, turns both strips off. (1 mark)

Question 39 | 4 marks

special-39-02-q39

A student claims that a wire whose resistance changes must have changed its material. An intact uniform wire initially has length L , cross-sectional area A and resistance $10\ \Omega$. It is stretched uniformly to length $2L$ without loss of material, so its volume LA stays constant. At the original temperature the stretched wire draws $0.10\ \text{A}$ from an ideal $4.0\ \text{V}$ source. It is then cut into two equal pieces, and those pieces are connected in parallel across the same source. Ignore contact resistance. (i) Use volume conservation to find the stretched area and, under the hypothesis that resistivity is unchanged, predict its resistance. (1 mark) (ii) Use the measured current to assess whether a material change is needed to explain the observation. (1 mark) (iii) Find the equivalent resistance after cutting and reconnecting, and compare the new source current with that of the original unstretched wire at $4.0\ \text{V}$. (2 marks)

Detailed solutions and marking guidance

Original model answers, numbered to match the question paper. These are not an official marking scheme.

Solution 1 — special-39-02-q01

Correct option: B

B. Electrons carry negative charge. Their net drift is opposite to the defined direction of conventional current, so they drift from Q towards P. Random electron motion does not remove this net drift. [1 mark]

Mark allocation

Choose B. Electrons carry negative charge. Their net drift is opposite to the defined direction of conventional current, so they drift from Q towards P. Random electron motion does not remove this net drift. [1 marks]

Solution 2 — special-39-02-q02

Correct option: C

C. The ideal source maintains its terminal potential difference even when the resistor path is open and no current flows through that path. Closing the switch permits current through the resistor but does not change the ideal source voltage, so the voltmeter reading remains the same. [1 mark]

Mark allocation

Choose C. The ideal source maintains its terminal potential difference even when the resistor path is open and no current flows through that path. Closing the switch permits current through the resistor but does not change the ideal source voltage, so the voltmeter reading remains the same. [1 marks]

Solution 3 — special-39-02-q03

Correct option: A

A. The incoming current is $90/30 = 3$ A. Branch currents add, so the other current is $3 - 0.80 = 2.20$ A. [1 mark]

Mark allocation

Choose A. The incoming current is $90/30 = 3$ A. Branch currents add, so the other current is $3 - 0.80 = 2.20$ A. [1 marks]

Solution 4 — special-39-02-q04

Correct option: D

D. K joins both ends of the $6\ \Omega$ resistor with negligible resistance, making its voltage and current zero. Only $3\ \Omega$ limits the source current: $9/3 = 3$ A. [1 mark]

Mark allocation

Choose D. K joins both ends of the $6\ \Omega$ resistor with negligible resistance, making its voltage and current zero. Only $3\ \Omega$ limits the source current: $9/3 = 3$ A. [1 marks]

Solution 5 — special-39-02-q05

Correct option: B

B. Its voltage remains 5 V and its current remains $5/10 = 0.50$ A; the source additionally supplies $5/15$ A to the new branch. [1 mark]

Mark allocation

Choose B. Its voltage remains 5 V and its current remains $5/10 = 0.50$ A; the source additionally supplies $5/15$ A to the new branch. [1 marks]

Solution 6 — special-39-02-q06

Correct option: C

C. Resistivity characterises the material at a specified temperature. The resistance of a particular wire also depends on its length and cross-sectional area through $R = \rho L/A$. Length and area describe specimens, not intrinsic material properties. [1 mark]

Mark allocation

Choose C. Resistivity characterises the material at a specified temperature. The resistance of a particular wire also depends on its length and cross-sectional area through $R = \rho L/A$. Length and area describe specimens, not intrinsic material properties. [1 marks]

Solution 7 — special-39-02-q07

Correct option: A

A. Area becomes one-quarter; $R \propto L/A$ gives $R_{\text{new}}/R_{\text{old}} = 2/(1/4) = 8$. [1 mark]

Mark allocation

Choose A. Area becomes one-quarter; $R \propto L/A$ gives $R_{\text{new}}/R_{\text{old}} = 2/(1/4) = 8$. [1 marks]

Solution 8 — special-39-02-q08

Correct option: D

D. Read the separate traces at 3.0 V: $I_X = 0.60$ A and $I_Y = 0.30$ A. In parallel both resistors have this voltage, so total current is $0.60 + 0.30 = 0.90$ A and the proposed trace fits. For series, $R_X = 3.0/0.60 = 5\ \Omega$ and $R_Y = 3.0/0.30 = 10\ \Omega$; total current would be $3.0/(5 + 10) = 0.20$ A, not 0.90 A. At constant resistance both network traces are straight through the origin. [1 mark]

Mark allocation

Choose D. Read the separate traces at 3.0 V: $I_X = 0.60$ A and $I_Y = 0.30$ A. In parallel both resistors have this voltage, so total current is $0.60 + 0.30 = 0.90$ A and the proposed trace fits. For series, $R_X = 3.0/0.60 = 5\ \Omega$ and $R_Y = 3.0/0.30 = 10\ \Omega$; total current would be $3.0/(5 + 10) = 0.20$ A, not 0.90 A. At constant resistance both network traces are straight through the origin. [1 marks]

Solution 9 — special-39-02-q09

Correct option: C

C. The shared current is $4/4 = 1$ A. Across $4 + 8 = 12\ \Omega$ the reading is $1 \times 12 = 12$ V. [1 mark]

Mark allocation

Choose C. The shared current is $4/4 = 1$ A. Across $4 + 8 = 12\ \Omega$ the reading is $1 \times 12 = 12$ V. [1 marks]

Solution 10 — special-39-02-q10

Correct option: B

B. It is energy used at 1000 W for 3600 s, equal to 3.6×10^6 J; it is not a power. [1 mark]

Mark allocation

Choose B. It is energy used at 1000 W for 3600 s, equal to 3.6×10^6 J; it is not a power. [1 marks]

Solution 11 — special-39-02-q11

Correct option: A

A. $H = V^2t/R$ at fixed voltage and time, so doubling R halves H. The current is not fixed. [1 mark]

Mark allocation

Choose A. $H = V^2t/R$ at fixed voltage and time, so doubling R halves H. The current is not fixed. [1 marks]

Solution 12 — special-39-02-q12

Correct option: D

D. The two intact branches remain across the same source voltage with unchanged resistance, so their currents remain unchanged; only source current falls. [1 mark]

Mark allocation

Choose D. The two intact branches remain across the same source voltage with unchanged resistance, so their currents remain unchanged; only source current falls. [1 marks]

Solution 13 — special-39-02-q13

Correct option: C

C. Each permits at most 1 A: $\sqrt{(2/2)} = \sqrt{(8/8)} = 1$. Thus $V_{\max} = 1 \times (2 + 8) = 10$ V. At 12 V both ratings would be exceeded. [1 mark]

Mark allocation

Choose C. Each permits at most 1 A: $\sqrt{(2/2)} = \sqrt{(8/8)} = 1$. Thus $V_{\max} = 1 \times (2 + 8) = 10$ V. At 12 V both ratings would be exceeded. [1 marks]

Solution 14 — special-39-02-q14

Correct option: B

B. $E = Pt = 0.050\text{ kW} \times 2\text{ h} = 0.10\text{ kWh}$. [1 mark]

Mark allocation

Choose B. $E = Pt = 0.050 \text{ kW} \times 2 \text{ h} = 0.10 \text{ kWh}$. [1 marks]

Solution 15 — special-39-02-q15

Correct option: A

A. V/I increases from 10Ω to 12Ω . The readings establish increased resistance but do not alone prove heating is the cause; temperature should be monitored or controlled. [1 mark]

Mark allocation

Choose A. V/I increases from 10Ω to 12Ω . The readings establish increased resistance but do not alone prove heating is the cause; temperature should be monitored or controlled. [1 marks]

Solution 16 — special-39-02-q16

Correct option: D

D. Their resistances are 24Ω and 12Ω . In series both carry the same current, so I^2R is larger in the 24Ω lamp: the lamp rated 6 W is brighter in this connection. [1 mark]

Mark allocation

Choose D. Their resistances are 24Ω and 12Ω . In series both carry the same current, so I^2R is larger in the 24Ω lamp: the lamp rated 6 W is brighter in this connection. [1 marks]

Solution 17 — special-39-02-q17

Correct option: A

A. A and R are true. With no branching or accumulation, the same charge per second must enter and leave, so R explains A. [1 mark]

Mark allocation

Choose A. A and R are true. With no branching or accumulation, the same charge per second must enter and leave, so R explains A. [1 marks]

Solution 18 — special-39-02-q18

Correct option: D

D. A is false: each piece has half the resistance, but unchanged resistivity. R is true. [1 mark]

Mark allocation

Choose D. A is false: each piece has half the resistance, but unchanged resistivity. R is true. [1 marks]

Solution 19 — special-39-02-q19

Correct option: B

B. Both are true, but the unit definition does not explain voltage equality. Equality follows because both resistors join the same two nodes. [1 mark]

Mark allocation

Choose B. Both are true, but the unit definition does not explain voltage equality. Equality follows because both resistors join the same two nodes. [1 marks]

Solution 20 — special-39-02-q20

Correct option: D

D. A is false: doubling R at fixed V halves P. R is true and gives that result directly. [1 mark]

Mark allocation

Choose D. A is false: doubling R at fixed V halves P. R is true and gives that result directly. [1 marks]

Solution 21 — special-39-02-q21

1. No charge accumulates, so $Q_Y = 64 - 24 = 40$ C. $I_Y = Q_Y/t = 40/40 = 1.0$ A. [1 mark]
2. $I_X = 24/40 = 0.60$ A, not 1.0 A. In parallel the voltages are equal, while branch currents depend on resistance and add to the 1.60 A supply current. [1 mark]

Mark allocation

No charge accumulates, so $Q_Y = 64 - 24 = 40$ C. $I_Y = Q_Y/t = 40/40 = 1.0$ A. [1 marks]

$I_X = 24/40 = 0.60$ A, not 1.0 A. In parallel the voltages are equal, while branch currents depend on resistance and add to the 1.60 A supply current. [1 marks]

Solution 22 — special-39-02-q22

1. A potential difference of 8 V means 8 J of energy is transferred per coulomb passing between the resistor's terminals. [1 mark]
2. $W = VQ = 8 \times 15 = 120$ J. [1 mark]

Mark allocation

A potential difference of 8 V means 8 J of energy is transferred per coulomb passing between the resistor's terminals. [1 marks]

$W = VQ = 8 \times 15 = 120$ J. [1 marks]

Solution 23 — special-39-02-q23

1. Subtract the positive zero offset: $I = 0.34 - 0.04 = 0.30$ A. $R = 3.6/0.30 = 12$ Ω . [1 mark]
2. The uncorrected current is too large. Dividing the same voltage by too large a current gives too small a resistance, so 10.6 Ω is an underestimate. [1 mark]

Mark allocation

Subtract the positive zero offset: $I = 0.34 - 0.04 = 0.30$ A. $R = 3.6/0.30 = 12$ Ω . [1 marks]

The uncorrected current is too large. Dividing the same voltage by too large a current gives too small a resistance, so 10.6 Ω is an underestimate. [1 marks]

Solution 24 — special-39-02-q24

1. Length, cross-sectional area and material all differ. A current/resistance difference could be due to any of them, so length has not been isolated. [1 mark]
2. Use two uniform wires of the same material and cross-sectional area but different measured

lengths. Keep their temperature the same, for example by taking brief readings and disconnecting between trials. Use the same test voltage. [1 mark]

Mark allocation

Length, cross-sectional area and material all differ. A current/resistance difference could be due to any of them, so length has not been isolated. [1 marks]

Use two uniform wires of the same material and cross-sectional area but different measured lengths. Keep their temperature the same, for example by taking brief readings and disconnecting between trials. Use the same test voltage. [1 marks]

Solution 25 — special-39-02-q25

1. $1 \text{ mm}^2 = 10^{-6} \text{ m}^2$, so $A = 0.30 \times 10^{-6} = 3.0 \times 10^{-7} \text{ m}^2$. [1 mark]

2. $\rho = RA/L = (2.0 \times 3.0 \times 10^{-7})/1.5 = 4.0 \times 10^{-7} \Omega \text{ m}$. [1 mark]

Mark allocation

$1 \text{ mm}^2 = 10^{-6} \text{ m}^2$, so $A = 0.30 \times 10^{-6} = 3.0 \times 10^{-7} \text{ m}^2$. [1 marks]

$\rho = RA/L = (2.0 \times 3.0 \times 10^{-7})/1.5 = 4.0 \times 10^{-7} \Omega \text{ m}$. [1 marks]

Solution 26 — special-39-02-q26

1. Equal currents mean equal charge per second enters and leaves. Charge is not being used up or accumulating in the heater. [1 mark]

2. The source supplies electrical energy, not a continuing consumption of charge. Across the heater energy is transferred as heat; a charge Q transfers VQ while the charge itself is conserved. [1 mark]

Mark allocation

Equal currents mean equal charge per second enters and leaves. Charge is not being used up or accumulating in the heater. [1 marks]

The source supplies electrical energy, not a continuing consumption of charge. Across the heater energy is transferred as heat; a charge Q transfers VQ while the charge itself is conserved. [1 marks]

Solution 27 — special-39-02-q27

1. In both drawings the 2Ω resistor joins A to N, while both 6Ω and 3Ω join N to B. There is a junction at N, so the three resistors are not all in series. [1 mark]

2. In each drawing 6Ω and 3Ω are parallel: $R_p = (6 \times 3)/(6 + 3) = 2 \Omega$. The 2Ω resistor is in series with this pair, so $R_{eq} = 2 + 2 = 4 \Omega$ in each. [1 mark]

3. Yes. Each source current is $I = 8/4 = 2 \text{ A}$. The two drawings have the same connections and terminal resistance despite their different shapes. [1 mark]

Mark allocation

In both drawings the 2Ω resistor joins A to N, while both 6Ω and 3Ω join N to B. There is a junction at N, so the three resistors are not all in series. [1 marks]

In each drawing 6Ω and 3Ω are parallel: $R_p = (6 \times 3)/(6 + 3) = 2 \Omega$. The 2Ω resistor is in series with this pair, so $R_{eq} = 2 + 2 = 4 \Omega$ in each. [1 marks]

Yes. Each source current is $I = 8/4 = 2$ A. The two drawings have the same connections and terminal resistance despite their different shapes. [1 marks]

Solution 28 — special-39-02-q28

- Each $12\ \Omega$ branch carries $6/12 = 0.50$ A; source current is $0.50 + 0.50 = 1.0$ A. [1 mark]
- Opening K_1 breaks only the Y branch. Source current becomes 0.50 A. X still spans the full 6 V and its current stays 0.50 A. [1 mark]
- Opening K_0 gives source current 0 A because K_0 is in the common series lead and breaks the supply path for both branches. K_1 controls only Y, whereas K_0 controls both. [1 mark]

Mark allocation

Each $12\ \Omega$ branch carries $6/12 = 0.50$ A; source current is $0.50 + 0.50 = 1.0$ A. [1 marks]

Opening K_1 breaks only the Y branch. Source current becomes 0.50 A. X still spans the full 6 V and its current stays 0.50 A. [1 marks]

Opening K_0 gives source current 0 A because K_0 is in the common series lead and breaks the supply path for both branches. K_1 controls only Y, whereas K_0 controls both. [1 marks]

Solution 29 — special-39-02-q29

- Since area is proportional to diameter squared, $R_X : R_Y : R_Z = L/d^2 : 2L/(2d)^2 : 4L/(2d)^2 = 1 : 1/2 : 1$, or $2 : 1 : 2$. [1 mark]
- X and Z have equal resistance and therefore equal current $I = V/R$ at the same voltage. Y carries twice either current. [1 mark]
- Yes. All have the same material and temperature, so their resistivity is the same. Different L/A changes resistance, not resistivity. [1 mark]

Mark allocation

Since area is proportional to diameter squared, $R_X : R_Y : R_Z = L/d^2 : 2L/(2d)^2 : 4L/(2d)^2 = 1 : 1/2 : 1$, or $2 : 1 : 2$. [1 marks]

X and Z have equal resistance and therefore equal current $I = V/R$ at the same voltage. Y carries twice either current. [1 marks]

Yes. All have the same material and temperature, so their resistivity is the same. Different L/A changes resistance, not resistivity. [1 marks]

Solution 30 — special-39-02-q30

- The ammeter is in the common lead before the junction, so it measures current through both R and the $6\ \Omega$ branch. $6.0/1.5 = 4\ \Omega$ is the parallel equivalent, not R. [1 mark]
- The $6\ \Omega$ branch carries $6/6 = 1$ A. R carries $1.5 - 1.0 = 0.50$ A. Thus $R = 6/0.50 = 12\ \Omega$. [1 mark]
- Move the ammeter into the R branch, in series with R, not into the other branch or across R. Replace its old common-lead position with an ideal wire. Leave the voltmeter across the two common nodes. [1 mark]

Mark allocation

The ammeter is in the common lead before the junction, so it measures current through both R and the $6\ \Omega$ branch. $6.0/1.5 = 4\ \Omega$ is the parallel equivalent, not R. [1 marks]

The $6\ \Omega$ branch carries $6/6 = 1\text{ A}$. R carries $1.5 - 1.0 = 0.50\text{ A}$. Thus $R = 6/0.50 = 12\ \Omega$. [1 marks]

Move the ammeter into the R branch, in series with R, not into the other branch or across R. Replace its old common-lead position with an ideal wire. Leave the voltmeter across the two common nodes. [1 marks]

Solution 31 — special-39-02-q31

1. In series the current and time are equal. $H = I^2Rt$ therefore gives $H_{3\Omega} : H_{6\Omega} = 3 : 6 = 1 : 2$. [1 mark]
2. In parallel the voltage and time are equal, while currents differ. $H = V^2t/R$ gives $H_{3\Omega} : H_{6\Omega} = (1/3) : (1/6) = 2 : 1$. [1 mark]
3. Joule's law: $H = I^2Rt$, with H in joules when I is in amperes, R in ohms and t in seconds. [1 mark]

Mark allocation

In series the current and time are equal. $H = I^2Rt$ therefore gives $H_{3\Omega} : H_{6\Omega} = 3 : 6 = 1 : 2$. [1 marks]

In parallel the voltage and time are equal, while currents differ. $H = V^2t/R$ gives $H_{3\Omega} : H_{6\Omega} = (1/3) : (1/6) = 2 : 1$. [1 marks]

Joule's law: $H = I^2Rt$, with H in joules when I is in amperes, R in ohms and t in seconds. [1 marks]

Solution 32 — special-39-02-q32

1. $EA = 0.030 \times 4 = 0.120\text{ kWh}$. $EB = (0.018 \times 3) + (0.042 \times 1) = 0.096\text{ kWh}$. [1 mark]
2. Only B meets the 0.10 kWh cap, with a margin of $0.100 - 0.096 = 0.004\text{ kWh}$. A exceeds it by 0.020 kWh . [1 mark]
3. The saving is $(0.120 - 0.096) \times ₹8 = ₹0.192$ (19.2 paise). Equivalently the costs are ₹0.960 and ₹0.768. [1 mark]

Mark allocation

$EA = 0.030 \times 4 = 0.120\text{ kWh}$. $EB = (0.018 \times 3) + (0.042 \times 1) = 0.096\text{ kWh}$. [1 marks]

Only B meets the 0.10 kWh cap, with a margin of $0.100 - 0.096 = 0.004\text{ kWh}$. A exceeds it by 0.020 kWh . [1 marks]

The saving is $(0.120 - 0.096) \times ₹8 = ₹0.192$ (19.2 paise). Equivalently the costs are ₹0.960 and ₹0.768. [1 marks]

Solution 33 — special-39-02-q33

1. The missing current is $3.5 - 2.5 = 1.0\text{ A}$. The intact branch currents would be $12/6 = 2\text{ A}$, $12/12 = 1\text{ A}$ and $12/24 = 0.5\text{ A}$; therefore the $12\ \Omega$ branch is open. [1 mark]
2. The $6\ \Omega$ branch still carries 2.0 A and the $24\ \Omega$ branch 0.50 A . Their sum is 2.50 A because they retain 12 V across them. [1 mark]
3. The voltmeter reads 12 V : the broken branch's terminals are still connected to opposite source terminals. An open path prevents current but does not require its terminal potential difference to be zero. [1 mark]

Mark allocation

The missing current is $3.5 - 2.5 = 1.0$ A. The intact branch currents would be $12/6 = 2$ A, $12/12 = 1$ A and $12/24 = 0.5$ A; therefore the $12\ \Omega$ branch is open. [1 marks]

The $6\ \Omega$ branch still carries 2.0 A and the $24\ \Omega$ branch 0.50 A. Their sum is 2.50 A because they retain 12 V across them. [1 marks]

The voltmeter reads 12 V: the broken branch's terminals are still connected to opposite source terminals. An open path prevents current but does not require its terminal potential difference to be zero. [1 marks]

Solution 34 — special-39-02-q34

1. Required $R_{eq} = V/I = 12/1.0 = 12\ \Omega$. [1 mark]
2. One design: connect two $12\ \Omega$ resistors in series in branch 1 and the other two in series in branch 2; connect these $24\ \Omega$ branches in parallel across the source. Each resistor is used without a bypass. Equivalently, connect two parallel pairs in series. Award 1 mark for a complete valid connection description or diagram. [1 mark]
3. For the first design $R_{eq} = (24 \times 24)/(24 + 24) = 12\ \Omega$, giving exactly 1.0 A. For the equivalent alternative, each parallel pair is $6\ \Omega$ and $6 + 6 = 12\ \Omega$. Award 1 mark for verification. [1 mark]
4. The current divides equally: each resistor carries 0.50 A and has 6 V across it. Its power is $(0.50)^2 \times 12 = 3.0$ W, below 4.0 W. All four therefore comply; total power is 12 W, matching VI. [1 mark]
5. In either valid design each resistor receives half the source voltage. Set $(V_{max}/2)^2/12 = 4$, giving $V_{max}^2 = 192$ and $V_{max} = 8\sqrt{3}\text{ V} \approx 13.86\text{ V}$. At this limit each resistor dissipates 4 W. [1 mark]

Mark allocation

Required $R_{eq} = V/I = 12/1.0 = 12\ \Omega$. [1 marks]

One design: connect two $12\ \Omega$ resistors in series in branch 1 and the other two in series in branch 2; connect these $24\ \Omega$ branches in parallel across the source. Each resistor is used without a bypass. Equivalently, connect two parallel pairs in series. Award 1 mark for a complete valid connection description or diagram. [1 marks]

For the first design $R_{eq} = (24 \times 24)/(24 + 24) = 12\ \Omega$, giving exactly 1.0 A. For the equivalent alternative, each parallel pair is $6\ \Omega$ and $6 + 6 = 12\ \Omega$. Award 1 mark for verification. [1 marks]

The current divides equally: each resistor carries 0.50 A and has 6 V across it. Its power is $(0.50)^2 \times 12 = 3.0$ W, below 4.0 W. All four therefore comply; total power is 12 W, matching VI. [1 marks]

In either valid design each resistor receives half the source voltage. Set $(V_{max}/2)^2/12 = 4$, giving $V_{max}^2 = 192$ and $V_{max} = 8\sqrt{3}\text{ V} \approx 13.86\text{ V}$. At this limit each resistor dissipates 4 W. [1 marks]

Solution 35 — special-39-02-q35

1. The $8\ \Omega$ resistor between Q and R is open. With zero current, each intact finite resistor has voltage $IR = 0$, explaining $V_{PQ} = V_{RS} = 0$. Thus P and Q are at the positive-terminal potential, while R and S are at the negative-terminal potential; the entire 12 V appears across the break. Award 1 for identification and 1 for this consistent explanation. [2 marks]

2. After repair $R_{eq} = 4 + 8 + 12 = 24 \, \Omega$; source current is $12/24 = 0.50 \, \text{A}$. The voltage across $8 \, \Omega$ is $0.50 \times 8 = 4.0 \, \text{V}$. [1 mark]

3. Test 2 gives $R_{eq} = 12/1.0 = 12 \, \Omega$. Compared with the intact $24 \, \Omega$ chain, $12 \, \Omega$ has been removed electrically; therefore the $12 \, \Omega$ resistor is bypassed. The ideal wire joins its terminals, so its voltage is $0 \, \text{V}$ and no current flows through that resistor itself. Award 1 for the equivalent-resistance deduction and 1 for the identified bypass with zero-voltage explanation. [2 marks]

Mark allocation

The $8 \, \Omega$ resistor between Q and R is open. With zero current, each intact finite resistor has voltage $IR = 0$, explaining $V_{PQ} = V_{RS} = 0$. Thus P and Q are at the positive-terminal potential, while R and S are at the negative-terminal potential; the entire $12 \, \text{V}$ appears across the break. Award 1 for identification and 1 for this consistent explanation. [2 marks]

After repair $R_{eq} = 4 + 8 + 12 = 24 \, \Omega$; source current is $12/24 = 0.50 \, \text{A}$. The voltage across $8 \, \Omega$ is $0.50 \times 8 = 4.0 \, \text{V}$. [1 marks]

Test 2 gives $R_{eq} = 12/1.0 = 12 \, \Omega$. Compared with the intact $24 \, \Omega$ chain, $12 \, \Omega$ has been removed electrically; therefore the $12 \, \Omega$ resistor is bypassed. The ideal wire joins its terminals, so its voltage is $0 \, \text{V}$ and no current flows through that resistor itself. Award 1 for the equivalent-resistance deduction and 1 for the identified bypass with zero-voltage explanation. [2 marks]

Solution 36 — special-39-02-q36

1. Mode F has $R = 12 \, \Omega$, so $P_F = V^2/R = 144/12 = 12 \, \text{W}$. [1 mark]
2. Mode S has $R = 24 \, \Omega$, so $P_S = 144/24 = 6 \, \text{W}$. [1 mark]
3. $t_F = 720/12 = 60 \, \text{s}$, while $t_S = 720/6 = 120 \, \text{s}$. Only F meets the $90 \, \text{s}$ deadline. [1 mark]
4. The claim is false under the stated loss-free transfer assumption: both require $720 \, \text{J}$ for a complete delivery. Lower power means a longer time, not less required energy. [1 mark]
5. $Q = W/V = 720/12 = 60 \, \text{C}$ for either complete delivery, although F delivers this in $60 \, \text{s}$ and S in $120 \, \text{s}$. This also follows from It : $1 \, \text{A} \times 60 \, \text{s}$ or $0.5 \, \text{A} \times 120 \, \text{s}$. [1 mark]

Mark allocation

Mode F has $R = 12 \, \Omega$, so $P_F = V^2/R = 144/12 = 12 \, \text{W}$. [1 marks]

Mode S has $R = 24 \, \Omega$, so $P_S = 144/24 = 6 \, \text{W}$. [1 marks]

$t_F = 720/12 = 60 \, \text{s}$, while $t_S = 720/6 = 120 \, \text{s}$. Only F meets the $90 \, \text{s}$ deadline. [1 marks]

The claim is false under the stated loss-free transfer assumption: both require $720 \, \text{J}$ for a complete delivery. Lower power means a longer time, not less required energy. [1 marks]

$Q = W/V = 720/12 = 60 \, \text{C}$ for either complete delivery, although F delivers this in $60 \, \text{s}$ and S in $120 \, \text{s}$. This also follows from It : $1 \, \text{A} \times 60 \, \text{s}$ or $0.5 \, \text{A} \times 120 \, \text{s}$. [1 marks]

Solution 37 — special-39-02-q37

1. Both readings give $V/I = 9 \, \Omega$. Two $18 \, \Omega$ resistors produce $9 \, \Omega$ in parallel, whereas series would give $36 \, \Omega$. They were therefore parallel. [1 mark]
2. The later equivalent resistance is $6/(1/3) = 18 \, \Omega$. One parallel branch has opened, leaving the single intact $18 \, \Omega$ branch carrying $1/3 \, \text{A}$. [1 mark]

3. Doubling the voltage to 12 V would indeed give $12/18 = 2/3$ A, equal to the former total current, but this current now all passes through one resistor. [1 mark]
4. Its power would be $12^2/18 = 8$ W, exceeding its 5 W rating. Reject the proposal even though it restores source current; replace the failed resistor or use a lower safe voltage instead. [1 mark]

Mark allocation

Both readings give $V/I = 9\ \Omega$. Two $18\ \Omega$ resistors produce $9\ \Omega$ in parallel, whereas series would give $36\ \Omega$. They were therefore parallel. [1 marks]

The later equivalent resistance is $6/(1/3) = 18\ \Omega$. One parallel branch has opened, leaving the single intact $18\ \Omega$ branch carrying $1/3$ A. [1 marks]

Doubling the voltage to 12 V would indeed give $12/18 = 2/3$ A, equal to the former total current, but this current now all passes through one resistor. [1 marks]

Its power would be $12^2/18 = 8$ W, exceeding its 5 W rating. Reject the proposal even though it restores source current; replace the failed resistor or use a lower safe voltage instead. [1 marks]

Solution 38 — special-39-02-q38

1. The current claim is false. A remains directly across 10 V, so its current stays $10/20 = 0.50$ A when B is switched off; source current falls, not A's current. [1 mark]
2. $P_A = 10^2/20 = 5$ W and $P_B = 10^2/50 = 2$ W. During 4 min with both on, $E_1 = (5 + 2) \times 240 = 1680$ J. [1 mark]
3. During 2 min with only A, $E_2 = 5 \times 120 = 600$ J. Total $E = 2280$ J, so the 1800 J log is incorrect by 480 J. [1 mark]
4. The master switch is in the common lead needed by both complete paths. Opening it disconnects the source from both branches. B's switch breaks only the B path. [1 mark]

Mark allocation

The current claim is false. A remains directly across 10 V, so its current stays $10/20 = 0.50$ A when B is switched off; source current falls, not A's current. [1 marks]

$P_A = 10^2/20 = 5$ W and $P_B = 10^2/50 = 2$ W. During 4 min with both on, $E_1 = (5 + 2) \times 240 = 1680$ J. [1 marks]

During 2 min with only A, $E_2 = 5 \times 120 = 600$ J. Total $E = 2280$ J, so the 1800 J log is incorrect by 480 J. [1 marks]

The master switch is in the common lead needed by both complete paths. Opening it disconnects the source from both branches. B's switch breaks only the B path. [1 marks]

Solution 39 — special-39-02-q39

1. Constant volume gives $L A = (2L) A_{\text{new}}$, so $A_{\text{new}} = A/2$. At unchanged resistivity $R_{\text{new}}/R_{\text{old}} = (2L/L)/(A_{\text{new}}/A) = 2/(1/2) = 4$, hence $R_{\text{new}} = 40\ \Omega$. [1 mark]
2. The measured resistance is $4.0/0.10 = 40\ \Omega$, agreeing with the unchanged-resistivity prediction. The geometry change is sufficient; the reading does not require or establish a material change. [1 mark]
3. Each half of the stretched wire has length L and area $A/2$, so each has resistance $20\ \Omega$. In

parallel their equivalent resistance is $20/2 = 10\ \Omega$. [1 mark]

4. New source current is $4.0/10 = 0.40\ \text{A}$. The original unstretched wire also drew $4.0/10 = 0.40\ \text{A}$, so the total source current is restored, although each new branch carries $0.20\ \text{A}$. [1 mark]

Mark allocation

Constant volume gives $LA = (2L)A_{\text{new}}$, so $A_{\text{new}} = A/2$. At unchanged resistivity $R_{\text{new}}/R_{\text{old}} = (2L/L)/(A_{\text{new}}/A) = 2/(1/2) = 4$, hence $R_{\text{new}} = 40\ \Omega$. [1 marks]

The measured resistance is $4.0/0.10 = 40\ \Omega$, agreeing with the unchanged-resistivity prediction. The geometry change is sufficient; the reading does not require or establish a material change. [1 marks]

Each half of the stretched wire has length L and area $A/2$, so each has resistance $20\ \Omega$. In parallel their equivalent resistance is $20/2 = 10\ \Omega$. [1 marks]

New source current is $4.0/10 = 0.40\ \text{A}$. The original unstretched wire also drew $4.0/10 = 0.40\ \text{A}$, so the total source current is restored, although each new branch carries $0.20\ \text{A}$. [1 marks]

Source-pattern and rationale appendix

These questions are original. References identify supplied source patterns and curriculum connections, not official provenance for these questions or predictions of future examinations. No missing references or rationale have been inferred.

Question 1 — special-39-02-q01

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.1, printed pp.171-172

Competency classification: No

Competency marks: 0

Competency rationale: Foundational recall of the opposite directions of conventional current and electron drift; zero competency marks.

Rationale tags: foundational

Question 2 — special-39-02-q02

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.1, printed pp.171-172

NCERT Ch.11 §11.2, printed p.173

Competency classification: No

Competency marks: 0

Competency rationale: Foundational distinction between source potential difference and a closed conducting path; no numerical substitution and zero competency marks.

Rationale tags: foundational

Question 3 — special-39-02-q03

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4-11.5, printed pp.182-184

NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185-186

Competency classification: No

Competency marks: 0

Competency rationale: Conservative audit: routine current conversion followed by subtraction; no separate competency credit.

Rationale tags: foundational

Question 4 — special-39-02-q04

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4-11.5, printed pp.182-184

NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185-186

Competency classification: Yes

Competency marks: 1

Competency rationale: The wire topology must be interpreted before any resistance calculation; the bypass is not a third resistive branch.

Rationale tags: interpretation
reasoned-selection

Question 5 — special-39-02-q05

Source patterns: sqp_2020-21_q9_main
ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 1

Competency rationale: Distinguish unchanged branch voltage from changed total source current.

Rationale tags: interpretation
reasoned-selection

Question 6 — special-39-02-q06

Source patterns: SQP_2024-25_q32
ncert-11

NCERT references: NCERT Ch.11 §11.5, printed pp.178–179

Competency classification: No

Competency marks: 0

Competency rationale: Foundational classification of resistivity versus resistance and dimensions; zero competency marks.

Rationale tags: foundational

Question 7 — special-39-02-q07

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179

Competency classification: Yes

Competency marks: 1

Competency rationale: Account for the squared diameter dependence while combining two geometry changes.

Rationale tags: interpretation
reasoned-selection

Question 8 — special-39-02-q08

Source patterns: cbse_sqp_2017-18_electricity_q26
ncert-11

NCERT references: NCERT Ch.11 §11.4, Activity 11.1, printed pp.175–176
NCERT Ch.11 §11.6.1, printed pp.183–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 1

Competency rationale: One competency mark for extracting two currents from the supplied graph and testing competing series/parallel connection models against a proposed total-current trace; not a material-ratio

inference or an isolated supplied-value substitution.

Rationale tags: graph-evidence
connection-model-comparison

Question 9 — special-39-02-q09

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: No

Competency marks: 0

Competency rationale: Conservative audit: standard series-current and total-voltage calculation; changing voltmeter endpoints alone is insufficient for competency credit.

Rationale tags: foundational

Question 10 — special-39-02-q10

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.8, printed pp.191–192

Competency classification: No

Competency marks: 0

Competency rationale: Foundational meaning and conversion of an energy unit.

Rationale tags: foundational

Question 11 — special-39-02-q11

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.7, printed pp.188–189

Competency classification: No

Competency marks: 0

Competency rationale: Conservative audit: direct inverse proportionality at stated fixed voltage and time; foundational.

Rationale tags: foundational

Question 12 — special-39-02-q12

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: No

Competency marks: 0

Competency rationale: Conservative audit: direct recall of ideal parallel-branch independence; foundational despite fault wording.

Rationale tags: foundational

Question 13 — special-39-02-q13

Source patterns: cbse_sqp_2017-18_electricity_q7
ncert-11

NCERT references: NCERT Ch.11 §11.8, printed pp.191-192

Competency classification: Yes

Competency marks: 1

Competency rationale: Find the limiting current for each component rather than summing ratings without checking distribution.

Rationale tags: interpretation
reasoned-selection

Question 14 — special-39-02-q14

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.8, printed pp.191-192

Competency classification: No

Competency marks: 0

Competency rationale: Single power-time substitution and unit conversion.

Rationale tags: foundational

Question 15 — special-39-02-q15

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.4, printed p.176
NCERT Ch.11 §11.5, printed p.179

Competency classification: Yes

Competency marks: 1

Competency rationale: Distinguish a measured resistance change from an unproved explanation of its cause.

Rationale tags: interpretation
reasoned-selection

Question 16 — special-39-02-q16

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4-11.5, printed pp.182-184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185-186
NCERT Ch.11 §11.8, printed pp.191-192

Competency classification: Yes

Competency marks: 1

Competency rationale: Convert ratings to resistances and then compare powers under the new common-current condition.

Rationale tags: interpretation
reasoned-selection

Question 17 — special-39-02-q17

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184

NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: No

Competency marks: 0

Competency rationale: Direct chapter principle and its standard explanation; conservatively foundational.

Rationale tags: foundational

Question 18 — special-39-02-q18

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179

Competency classification: No

Competency marks: 0

Competency rationale: Conservative audit: textbook distinction between resistance and resistivity; foundational A/R item.

Rationale tags: foundational

Question 19 — special-39-02-q19

Source patterns: sqp_2020-21_q9_main
ncert-11

NCERT references: NCERT Ch.11 §11.2, printed p.173

NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184

NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: No

Competency marks: 0

Competency rationale: Recognize two textbook facts and an explanatory mismatch; no conservative competency credit.

Rationale tags: foundational

Question 20 — special-39-02-q20

Source patterns: pyq-2022-4f106d54aeecf637b6b663fc
ncert-11

NCERT references: NCERT Ch.11 §11.8, printed pp.191–192

Competency classification: No

Competency marks: 0

Competency rationale: Conservative audit: the supplied reason gives the formula and assertion needs only direct inverse proportionality; foundational.

Rationale tags: foundational

Question 21 — special-39-02-q21

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.1, printed pp.171–172

NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184

NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 2

Competency rationale: Infer missing branch flow from simultaneous charge records and refute equality of parallel currents.

Rationale tags: analysis
evidence-based-reasoning

Question 22 — special-39-02-q22

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.2, printed p.173

Competency classification: No

Competency marks: 0

Competency rationale: Definition and direct calculation; neither subpart is counted as competency.

Rationale tags: foundational

Question 23 — special-39-02-q23

Source patterns: pyq-2018-7253f82c5332b612f2b92314
ncert-11

NCERT references: NCERT Ch.11 §11.4, Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 2

Competency rationale: Correct systematic error and determine the direction of its effect on an inferred resistance.

Rationale tags: analysis
evidence-based-reasoning

Question 24 — special-39-02-q24

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179

Competency classification: Yes

Competency marks: 2

Competency rationale: Identify confounding variables and redesign a causal comparison.

Rationale tags: analysis
evidence-based-reasoning

Question 25 — special-39-02-q25

Source patterns: SQP_2024-25_q32
pyq-2022-cfec2ef8067dd02793a5ebf2
ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179

Competency classification: No

Competency marks: 0

Competency rationale: Routine unit conversion and substitution; classified foundational.

Rationale tags: foundational

Question 26 — special-39-02-q26

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.1, printed pp.171–172
NCERT Ch.11 §11.7, printed p.188
NCERT Ch.11 §11.8, More to Know, printed p.191

Competency classification: Yes

Competency marks: 1

Competency rationale: Only (i) receives 1 mark for using equal-current evidence to reject charge consumption. Part (ii) is the standard textbook energy-transfer explanation and is foundational.

Rationale tags: analysis
evidence-based-reasoning

Question 27 — special-39-02-q27

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 3

Competency rationale: Requires extracting a node-to-component map from two visually different layouts, then testing a topology claim.

Rationale tags: analysis
evidence-based-reasoning

Question 28 — special-39-02-q28

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 3

Competency rationale: The figure is essential for tracing which closed paths survive each switch action.

Rationale tags: analysis
evidence-based-reasoning

Question 29 — special-39-02-q29

Source patterns: SQP_2024-25_q32
ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179

Competency classification: Yes

Competency marks: 2

Competency rationale: Parts (i) and (ii) receive 2 marks for diameter-squared scaling and selecting an equal-current pair. Part (iii) repeats the same-material/same-temperature resistivity rule and is foundational.

Rationale tags: analysis
evidence-based-reasoning

Question 30 — special-39-02-q30

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.1, printed p.172
NCERT Ch.11 §11.2, printed p.173
NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 3

Competency rationale: Read the ammeter's position relative to a junction, subtract an unintended branch current, and repair the measurement design.

Rationale tags: analysis
evidence-based-reasoning

Question 31 — special-39-02-q31

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.7, printed pp.188–189
NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 2

Competency rationale: Only (i) and (ii) count: controlled-current versus controlled-voltage comparison. Formula recall in (iii) is foundational.

Rationale tags: analysis
evidence-based-reasoning

Question 32 — special-39-02-q32

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 1

Competency rationale: Only part (ii) receives 1 mark for the feasibility and margin decision. Energy calculations and multiplying the energy saving by the tariff are foundational arithmetic.

Rationale tags: analysis
evidence-based-reasoning

Question 33 — special-39-02-q33

Source patterns: pyq-2023-84af9d4edc02297aecab4a0f
ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 2

Competency rationale: Parts (i) and (iii) receive 2 marks for fault identification and interpreting nonzero voltage across an open path. Part (ii) is routine intact-branch current calculation.

Rationale tags: analysis
evidence-based-reasoning

Question 34 — special-39-02-q34

Source patterns: cbse_sqp_2017-18_electricity_q7
sqp_2020-21_q26
ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186
NCERT Ch.11 §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 4

Competency rationale: Part (i) is foundational. Parts (ii)–(iv) synthesize a topology, check actual power distribution and infer the limiting voltage; only these four marks count.

Rationale tags: design-or-diagnosis
multi-step-reasoning

Question 35 — special-39-02-q35

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 4

Competency rationale: Parts (i) and (iii) receive 4 marks for open-versus-bypass diagnosis from evidence. Part (ii), the repaired series calculation, is foundational.

Rationale tags: design-or-diagnosis
multi-step-reasoning

Question 36 — special-39-02-q36

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.7, printed pp.188–189
NCERT Ch.11 §11.8, printed pp.191–192
NCERT Ch.11 §11.2, printed p.173

Competency classification: Yes

Competency marks: 2

Competency rationale: Only the deadline comparison and rejection of the lower-power/less-energy claim receive 2 marks. The two power calculations and $Q=W/V$ substitution are foundational.

Rationale tags: design-or-diagnosis
multi-step-reasoning

Question 37 — special-39-02-q37

Source patterns: cbse_sqp_2017-18_electricity_q7
ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186
NCERT Ch.11 §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 4

Competency rationale: The initial table identifies hidden topology, the changed reading identifies its surviving path, and the proposed workaround must be tested against a component rating.

Rationale tags: design-or-diagnosis
multi-step-reasoning

Question 38 — special-39-02-q38

Source patterns: sqp_2020-21_q9_main
ncert-11

NCERT references: NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186
NCERT Ch.11 §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 3

Competency rationale: Parts (i), the second mark of (ii), and (iii) receive 3 marks for branch independence, reconciling the interval log and master-switch reasoning. The first energy-calculation mark is foundational.

Rationale tags: design-or-diagnosis
multi-step-reasoning

Question 39 — special-39-02-q39

Source patterns: SQP_2024-25_q32
ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179
NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 3

Competency rationale: Parts (i), (ii), and the reconnection-resistance mark in (iii) receive 3 marks for conserved-volume geometry, hypothesis/evidence comparison and topology change. The final current substitution/comparison is conservatively foundational.

Rationale tags: design-or-diagnosis
multi-step-reasoning

Source reference details

id: SQP_2024-25_q32

title: CBSE Class X Science (086) Sample Question Paper 2024-25 (English)

url: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

source_role: design-pattern

local_source_record: research/sqp/question_bank.json

question_number: 32

session: 2024-25

exam_type: SQP

original_section: Section-C

language: en

pages: 9

page: 9

bank_verified: Yes

bank_answer_verified: Yes

bank_answer_source_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: cbse_sqp_2017-18_electricity_q26

title: Sample Question Paper 2017-18 — Science — Class - X

url: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_SQP.pdf

source_role: design-pattern

local_source_record: research/sqp/question_bank.json

question_number: 26

session: 2017-18

exam_type: SQP

original_section: SECTION - B

language: en

pages: 3

4

page: 3

bank_verified: Yes

bank_answer_verified: No

bank_answer_source_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: cbse_sqp_2017-18_electricity_q7

title: Sample Question Paper 2017-18 — Science — Class - X

url: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_SQP.pdf

source_role: design-pattern

local_source_record: research/sqp/question_bank.json

question_number: 7

session: 2017-18

exam_type: SQP

original_section: SECTION - A

language: en

pages: 1

page: 1

bank_verified: Yes

bank_answer_verified: Yes

bank_answer_source_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: ncert-11
title: NCERT Science, Class X, Chapter 11: Electricity
url: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>
source_role: conceptual-scope
local_source_record: research/sqp/archive/jesc111.txt
notes: Conceptual scope and activity reference only; not provenance of original questions. No question-bank verification flags are inferred.

id: pyq-2018-7253f82c5332b612f2b92314
title: CBSE — Class X Science — 2018 — 31/B — main
url: https://www.cbse.gov.in/curric~1/qpms2018/qp_10/Science.zip
source_role: design-pattern
local_source_record: research/pyq/question_bank.json
question_number: 27
year: 2018
session: main
exam_type: PYQ
paper_code: 31/B
language: English
source_title_generated: Yes
session_inferred: Yes
original_section: SECTION-B
pages: 15
page: 15
bank_verified: Yes
bank_answer_verified: No
bank_official_source_verified: Yes
bank_text_verified: Yes
bank_visual_verified: Yes
bank_year_verified: Yes
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: pyq-2022-4f106d54aeecf637b6b663fc
title: CBSE — Class X Science — 2022 — compartment — 31/6/1
url: <https://www.cbse.gov.in/cbsenew/question-paper/2022-COMPTT/X/SCIENCE.zip>
source_role: design-pattern
local_source_record: research/pyq/question_bank.json
question_number: 12
year: 2022
session: compartment
exam_type: PYQ
paper_code: 31/6/1
language: English
source_title_generated: Yes
original_section: SECTION B
pages: 11
page: 11
bank_verified: Yes
bank_answer_verified: No
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: pyq-2022-cfec2ef8067dd02793a5ebf2
title: CBSE — Class X Science — 2022 — main — 31/1/2
url: <https://www.cbse.gov.in/cbsenew/question-paper/2022/X/Scince.zip>
source_role: design-pattern
local_source_record: research/pyq/question_bank.json

question_number: 10
year: 2022
session: main
exam_type: PYQ
paper_code: 31/1/2
term: II
language: English
source_title_generated: Yes
original_section: SECTION B
pages: 7
page: 7
bank_verified: Yes
bank_answer_verified: No
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.
id: pyq-2023-84af9d4edc02297aecab4a0f
title: CBSE — Class X Science — 2023 — main — 31/5/2
url: <https://www.cbse.gov.in/cbsenew/question-paper/2023/X/SCIENCE.zip>
source_role: design-pattern
local_source_record: research/pyq/question_bank.json
question_number: 34
year: 2023
session: main
exam_type: PYQ
paper_code: 31/5/2
language: English
source_title_generated: Yes
original_section: SECTION D
pages: 19
page: 19
bank_verified: Yes
bank_answer_verified: No
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.
id: sqp_2020-21_q26
title: CBSE Class X Science (086) Sample Question Paper 2020-21
url: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-SQP.pdf
source_role: design-pattern
local_source_record: research/sqp/question_bank.json
question_number: 26
session: 2020-21
exam_type: SQP
original_section: Section B
language: en
pages: 12
page: 12
bank_verified: Yes
bank_answer_verified: Yes
bank_answer_source_verified: Yes
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.
id: sqp_2020-21_q9_main
title: CBSE Class X Science (086) Sample Question Paper 2020-21
url: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-SQP.pdf
source_role: design-pattern

local_source_record: research/sqp/question_bank.json

question_number: 9

session: 2020-21

exam_type: SQP

original_section: Section A

language: en

pages: 2

page: 2

bank_verified: Yes

bank_answer_verified: Yes

bank_answer_source_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.